Figure 1: Fundamentals of Big Data

data gets added into the databases of social media site Facebook daily.

Types of Big Data

Big Data can be classified into the following three main categories.

1. **Structured:** Data that can be stocked, approached and refined in the form of a fixed data format is termed as structured data. With time, computer science has been able to develop methods for running with such data and also deriving value out of it. Nevertheless, these days, we are anticipating issues related to the sheer volume of such data, which is turning into zettabytes (1 billion terabytes equals 1 zettabyte).
2. **Unstructured:** Data in an unmapped form is known as unstructured data. Large volumes of unstructured data pose many challenges in terms of how to derive value out of it. For example, a heterogeneous data source, incorporating a collection of simple text files, pictures, audio as well as video recordings, will be difficult to analyse. These days, organisations have an abundance of data available to them, but unfortunately they don't know how to extract value out of it since this data is in an unprocessed form.
3. **Semi-structured:** This can comprise both forms of data. Also, we can consider semi-structured data as a structure in form, but in reality, the data itself is not defined, e.g., data depicted in an XML file.

The four Vs of Big Data

Some of the common characteristics of Big Data are depicted in Figure 2.

1. **Volume:** The volume of data is an important factor in deciding on its value. Hence, volume is one property that needs to be considered while handling Big Data.
2. **Variety:** This refers to assorted data sources and the nature of data, both structured and unstructured. Previously, spreadsheets and databases were the only origins of data considered in most of the practical applications. But these days, data in the form of e-mails, pictures, recordings, monitoring devices, etc, are also being considered in investigation applications.
3. **Velocity:** This term refers to how swiftly data is generated. How fast the data is created and refined to meet a particular need, determines its real potential. The velocity of Big Data is the rate at which data flows from sources like business procedures, application logs, websites, etc. The speed at which Big Data flows is very high and virtually non-stop.
4. **Veracity:** This refers to the incompatibility between the various formats that the data is being generated in, thus constraining the process of mining or managing the data profitably.

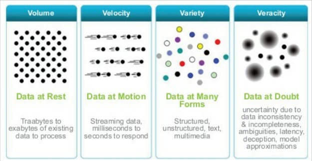

Figure 2: Characteristics of Big Data

Big Data architecture

Big Data architecture comprises consistent, scalable and completely computerised data pipelines. The skillset needed to build such infrastructure requires a deep knowledge of every layer in the heap, starting with a cluster design to setting up the top chain responsible for processing the data. Figure 3 shows the complexity of the stack, along with how data pipeline engineering touches every part of it.

In this figure, the data pipelines collect raw data and transform it into something of value. Meanwhile, the Big Data engineer has to plan what happens to the data, the way it is stored in the cluster, how access is approved internally, what equipment to use for processing the data, and finally, the mode of providing access to the outside world. Those who design and implement this architecture are referred to as Big Data engineers.

Big Data technologies

As we know, the subject of Big Data is very broad and permeates many new technology developments. Here is an overview of some of the technologies that help users monetise Big Data.

1. **MapReduce:** This allows job implementation, with scalability crossing thousands of servers.
   - ***Map:*** Input dataset transforms into a different set of values.
   - ***Reduce:*** Many outputs of the Map task are united to form a reduced set of values.